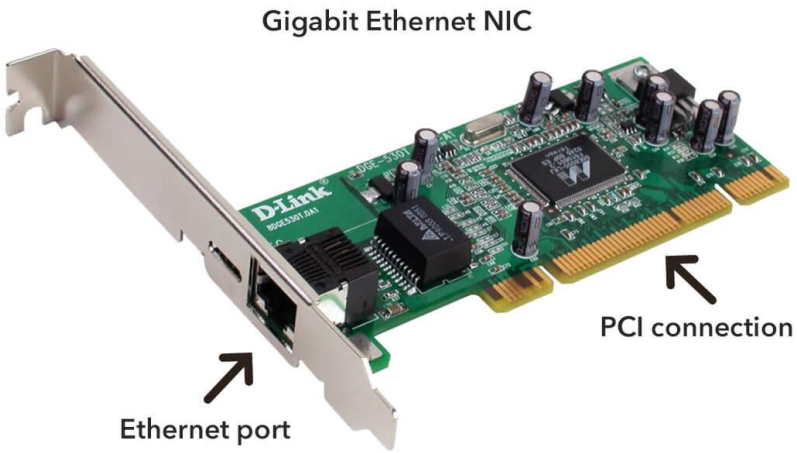
1.Study of Network Devices in detail and connect the computers in LAN

A Local Area Network (LAN) connects computers and devices within a limited geographic area, such as an office, school, or building, to share resources like files, printers, and the internet. The basic required components that computers are connected to using these devices.

1. Network Interface Card (NIC)

- Every computer or device must have an NIC to connect to a LAN. It assigns a unique MAC address to each device, allowing identification and data transfer.
- NICs can be wired (Ethernet) or wireless (Wi-Fi).



2. Switch

- Switches are central devices that connect multiple computers/devices. They use MAC addresses to forward data only to the target device, reducing congestion and collisions compared to hubs.
- In most modern LANs, switches are used for connecting devices on the same network efficiently.



3. Router

- Routers connect the LAN to other networks, such as the internet. They assign IP addresses to devices and determine the best path to send data.

- In a typical setup, a router sits between the LAN and the internet service provider.





4. Hub (Legacy)

- Hubs also connect multiple devices, but they send incoming data to all ports, regardless of destination. This can cause network inefficiency and security risks. Hubs are largely replaced by switches in modern LANs.



5. Wireless Access Point (WAP)

- WAPs allow wireless devices (laptops, smartphones) to connect to the LAN using Wi-Fi.
- They are commonly used in networks with mobile devices or where wiring is impractical.

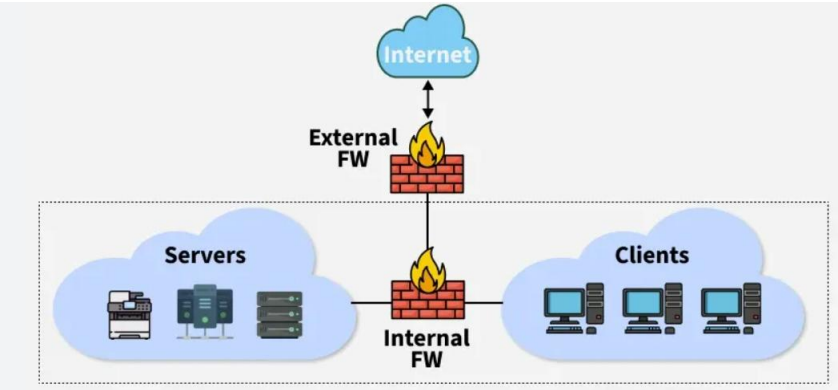


6. Cables and Connectors

- Ethernet cables (like Cat5e, Cat6) physically connect devices to switches/routers. These provide the backbone for wired LANs.
- For wireless, the connection is established through radio signals, with the WAP as the central point.

7. Additional Components

- Servers: Manage shared resources.
- Printers/scanners: Connected as network resources.
- Firewalls: Maintain security by blocking unauthorized access.



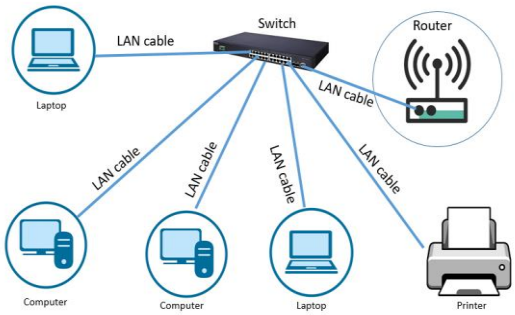
Step-by-Step Wired LAN Setup

1. Install NICs: Make sure all computers have NICs.
2. Connect Devices: Plug Ethernet cables from each computer into the ports on the switch.
3. Connect the Switch to the Router: Use another Ethernet cable to connect the switch to the router (for internet access).
4. Power On Devices: Turn on all computers and networking equipment.
5. Configure IP Addresses: Usually, the router uses DHCP to automatically assign IP addresses to all connected devices.
6. Test Connectivity: Check that devices can communicate by pinging each other's IP addresses or sharing files.

Step-by-Step Wireless LAN Setup

1. Install Wireless NICs: Make sure wireless devices have working Wi-Fi adapters.
2. Configure Access Point: Connect and configure the WAP with your desired settings (SSID, password).
3. Connect Devices Wirelessly: Search for the network on each device and enter the credentials to connect.
4. Router Integration: The WAP may connect to the router, either using a cable or wirelessly, for internet connectivity.

Sample LAN Diagram



Local Area Network

